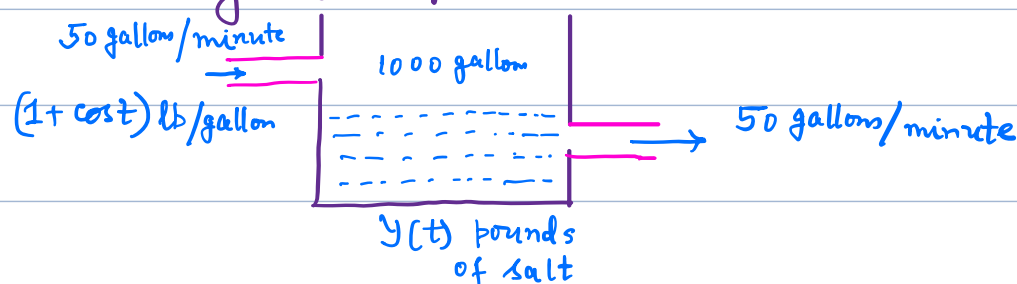


MTH 204 : Lecture 04

Ex: A tank contains 1000 gallons of water in which 200 pound salt is dissolved. 50 gallon of brine, each gallon containing $(1 + \cos t)$ pound of dissolved salt, runs into the tank per minute.

The mixture, kept uniform by stirring, runs out at the same rate. Find the amount of salt in the tank at any time.



Let $y(t)$ be the amount of salt at any time t .

Then $\frac{dy}{dt} = \text{salt inflow rate} - \text{salt outflow rate}$

$$\Rightarrow \frac{dy}{dt} = 50(1 + \cos t) - \frac{y}{1000} \times 50 = 50(1 + \cos t) - .05y$$

$$\Rightarrow y' + .05y = 50(1 + \cos t)$$

$$\text{The I.F.} = e^{\int .05 dy} = e^{.05t}$$

Then the solution is

$$y = e^{-.05t} \left[\int e^{.05t} 50(1 + \cos t) dt + C \right]$$

$$\Rightarrow y = e^{-.05t} \left[50 \int e^{.05t} (1 + \cos t) dt + C \right] \dots\dots\dots ①$$

$$\begin{aligned}\text{Now } \int e^{.05t} (1 + \cos t) dt &= \int e^{.05t} dt + \int e^{.05t} \cos t dt \\ &= \frac{e^{.05t}}{.05} + \int e^{.05t} \cos t dt\end{aligned}$$

$$\begin{aligned}\text{Let } I &= \int e^{.05t} \cos t dt = \frac{e^{.05t}}{.05} \cos t - \int (-\sin t) \frac{e^{.05t}}{.05} dt \\ &= \frac{e^{.05t}}{.05} \cos t + \frac{1}{.05} \int e^{.05t} \sin t dt \\ &= \frac{e^{.05t}}{.05} \cos t + \frac{1}{.05} \left[\frac{e^{.05t}}{.05} \sin t - \int \frac{e^{.05t}}{.05} \cos t dt \right] \\ &= \frac{e^{.05t}}{.05} \cos t + \frac{e^{.05t}}{(.05)^2} \sin t - \frac{1}{(.05)^2} \int e^{.05t} \cos t dt\end{aligned}$$

$$\Rightarrow I + \frac{I}{(.05)^2} = \frac{e^{.05t}}{.05} \cos t + \frac{e^{.05t}}{(.05)^2} \sin t$$

$$\Rightarrow I \left[\frac{(.05)^2 + 1}{(.05)^2} \right] = \frac{e^{.05t}}{.05} \cos t + \frac{e^{.05t}}{(.05)^2} \sin t$$

$$\Rightarrow I = \frac{.05}{(.05)^2 + 1} e^{.05t} \cos t + \frac{1}{(.05)^2 + 1} e^{.05t} \sin t$$

$$\begin{aligned}\Rightarrow 50I &= \frac{2.5}{(.05)^2 + 1} e^{.05t} \cos t + \frac{50}{(.05)^2 + 1} e^{.05t} \sin t \\ &= 2.494 e^{.05t} \cos t + 49.88 e^{.05t} \sin t\end{aligned}$$

Now from ①,

$$y(t) = e^{-.05t} \left[\frac{50}{.05} e^{.05t} + 2.494 e^{.05t} \cos t + 49.88 e^{.05t} \sin t + C \right]$$

$$\Rightarrow y(t) = e^{-.05t} \left[1000e^{.05t} + 2.494e^{.05t} \cos t + 49.88e^{.05t} \sin t + C \right]$$

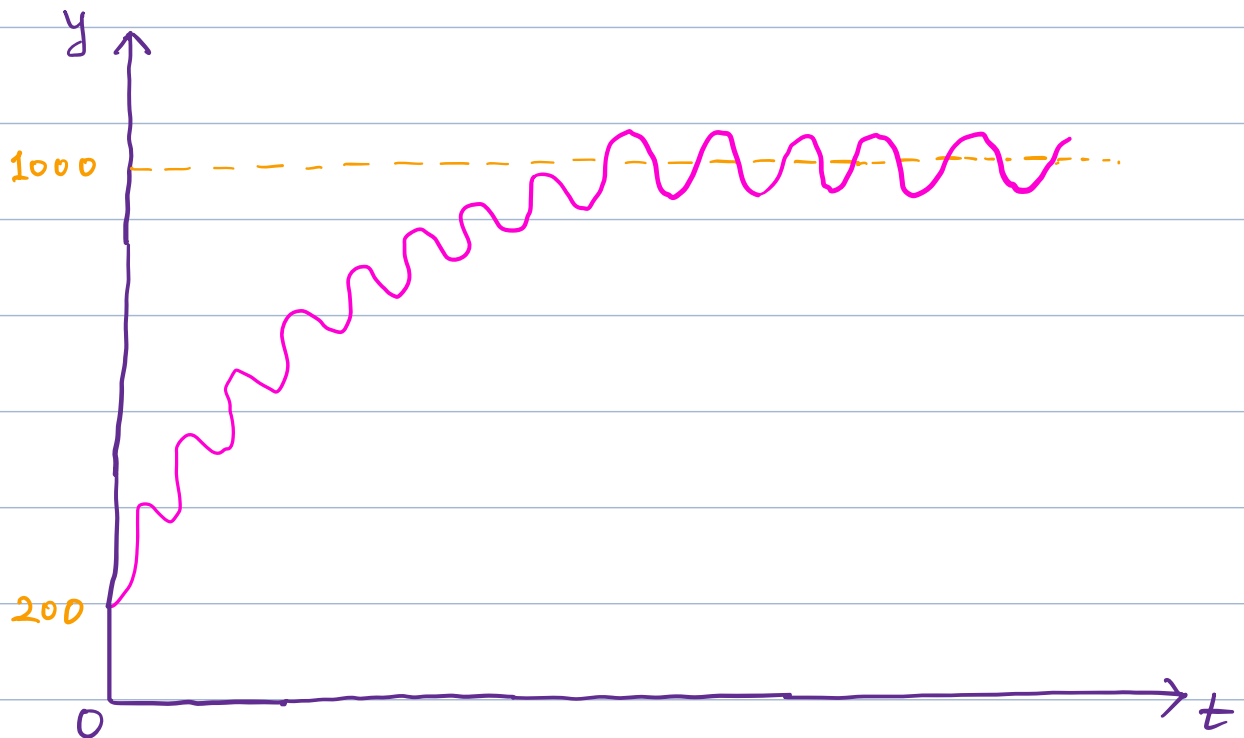
$$\Rightarrow y(t) = 1000 + 2.494 \cos t + 49.88 \sin t + C e^{-.05t}$$

$$\text{Now } y(0) = 200 \Rightarrow 200 = 1000 + 2.494 + C$$

$$\Rightarrow C = -802.494 = -802.5$$

Hence $y(t) = 1000 + 2.494 \cos t + 49.88 \sin t - 802.5 e^{-.05t}$

Thus $y(t)$ increases (as t increases) but keeps oscillating about 1000 as the limit of the mean value.



Ex: Assume that the level of a certain hormone in the blood of a patient varies with time. Suppose that the time rate of change is the difference between a sinusoidal input of a 24-hour period from the thyroid gland and a continuous removal rate proportional to the level present. Set up a model for the hormone level in the blood and find its general solution. Find the particular solution satisfying a suitable initial condition.

Let $y(t)$ be the level of the hormone in blood at time t .

$$\text{Then } \frac{dy}{dt} = \underset{\substack{\text{sinusoidal} \\ \text{input}}}{\left(A + B \cos(\omega t) \right)} - \underset{\substack{\text{Proportional to } y(t) \\ \text{(i.e. } k y(t) \text{ where } k \text{ is a constant)}}}{k y(t)}$$

$$\text{So, } \frac{dy}{dt} = A + B \cos(\omega t) - k y(t)$$

$$\Rightarrow \frac{dy}{dt} + k y(t) = A + B \cos(\omega t)$$

$$\text{I.F.} = e^{\int k dt} = e^{kt}$$

$$\text{Hence } y(t) = e^{-kt} \left[\int (A + B \cos(\omega t)) e^{kt} dt + C \right]$$

$$= e^{-kt} \left[\frac{A e^{kt}}{k} + B e^{kt} \frac{k \cos(\omega t) + \omega \sin(\omega t)}{\omega^2 + k^2} + C \right]$$

$$\text{because } \int e^{kt} \cos(\omega t) dt$$

$$= \frac{e^{kt}}{k} \cos(\omega t) - \int -\omega \sin \omega t \frac{e^{kt}}{k} dt$$

$$= \frac{e^{kt}}{k} \cos \omega t + \frac{\omega}{k} \int \sin \omega t e^{kt} dt$$

$$= \frac{e^{kt}}{k} \cos \omega t + \frac{\omega}{k} \left[\frac{e^{kt}}{k} \sin \omega t - \int \omega \cos \omega t \frac{e^{kt}}{k} dt \right]$$

$$= \frac{e^{kt}}{k} \cos \omega t + \frac{\omega}{k^2} e^{kt} \sin \omega t - \frac{\omega^2}{k^2} \int e^{kt} \cos \omega t dt$$

$$\Rightarrow \left[1 + \frac{\omega^2}{k^2} \right] \int e^{kt} \cos(\omega t) dt$$

$$= \frac{1}{k^2} e^{kt} [k \cos(\omega t) + \omega \sin(\omega t)]$$

$$\Rightarrow \frac{(k^2 + \omega^2)}{k^2} \int e^{kt} \cos(\omega t) dt = \frac{e^{kt}}{k^2} [k \cos(\omega t) + \omega \sin(\omega t)]$$

$$\Rightarrow I = \int e^{kt} \cos(\omega t) dt = \frac{e^{kt}}{(k^2 + \omega^2)} [k \cos(\omega t) + \omega \sin(\omega t)]$$

$$\text{Hence } y(t) = e^{-kt} \left[\frac{A}{k} e^{kt} + \frac{B}{\omega^2 + k^2} e^{kt} (k \cos(\omega t) + \omega \sin(\omega t)) + C \right]$$

$$\Rightarrow y(t) = \frac{A}{k} + \frac{B}{\omega^2 + k^2} (k \cos(\omega t) + \omega \sin(\omega t)) + C e^{-kt}$$

Since the variation is every 24 hours,

$$\text{frequency } \omega = \frac{2\pi}{T} = \frac{2\pi}{24} = \frac{\pi}{12}$$

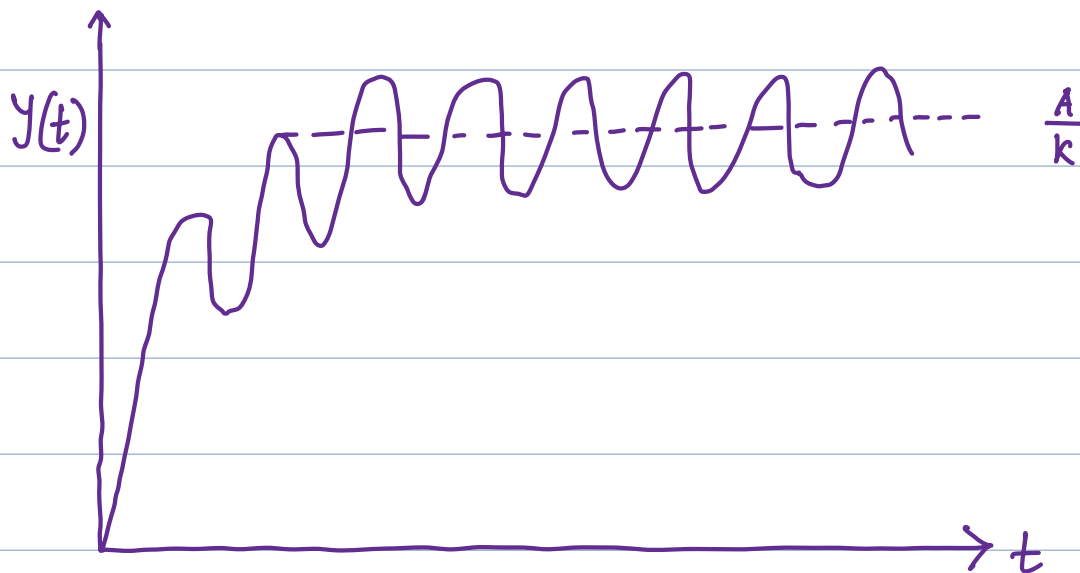
$$\text{Then } y = \frac{A}{k} + \frac{B}{\frac{\pi^2}{12^2} + k^2} \left(k \cos\left(\frac{\pi}{12} t\right) + \frac{\pi}{12} \sin\left(\frac{\pi}{12} t\right) \right) + C e^{-kt}$$

If we assume $y(0) = 0$, then

$$0 = \frac{A}{k} + \frac{Bk}{k^2 + \frac{\pi^2}{12^2}} + C \Rightarrow C = -\left(\frac{A}{k} + \frac{Bk}{k^2 + \frac{\pi^2}{12^2}} \right)$$

$$\text{Then } y = \frac{A}{k} (1 - e^{-kt}) + \frac{B}{k^2 + \frac{\pi^2}{12^2}} \left(k \cos\left(\frac{\pi}{12} t\right) + \frac{\pi}{12} \sin\left(\frac{\pi}{12} t\right) - k e^{-kt} \right)$$

In particular if $A = B = 1$ and $k = .05$, then the solution will have the following graphical form



Reduction to Linear ODEs:

Bernoulli Equation: $y' + p(x)y = g(x)y^a$

Question:

For what values of y , a Bernoulli Equation is linear? $\therefore a = 0$ and $a = 1$

Thus Bernoulli Equation is non-linear for $a \neq 0$ and $a \neq 1$

Let $u = y^{1-a}$

Then $u' = (1-a)y^{-a}y'$

$\Rightarrow u' = (1-a)y^{-a}(-p(x)y + g(x)y^a)$

$\Rightarrow u' = -(1-a)p(x)y^{1-a} + (1-a)g(x)$

$\Rightarrow u' = (1-a)(g(x) - p(x)u)$

$\Rightarrow \boxed{u' + (1-a)p \ u = (1-a)g} \rightarrow \text{Linear ODE}$

Ex: Solve: $y' + y = y^2$ ($a=2$), $y(0) = -1$

$$\text{Let } u = y^{1-a} = \frac{1}{y}$$

$$\text{Then } u' = (-1)y^{-1-1}y' = -\frac{y'}{y^2} \Rightarrow y' = -u'y^2 = -\frac{u'}{u^2}$$

$$\text{Then } y' + y = y^2 \Rightarrow -\frac{u'}{u^2} + \frac{1}{u} = \frac{1}{u^2} \Rightarrow -u' + u = 1$$

$$\Rightarrow u' - u = -1 \quad \dots\dots (1)$$

$$\text{I.F.} = \int e^{(-1)dx} = e^{-x}$$

So, the general solution of equation (1) is

$$u = e^x \left[\int (-e^{-x}) dx + C \right] = e^x \left[\frac{-e^{-x}}{-1} + C \right]$$

$$\Rightarrow u = 1 + ce^x \Rightarrow \frac{1}{y} = 1 + ce^x \Rightarrow y = \frac{1}{1+ce^x}$$

$$\text{Now } y(0) = -1 \Rightarrow -1 = \frac{1}{1+c} \Rightarrow c+1 = -1 \Rightarrow c = -2$$

Hence
$$y = \frac{1}{1-2e^x}$$

OR (By Method of Separation of Variables): $y' + y = y^2 \Rightarrow y' = y^2 - y = y(y-1)$

$$\Rightarrow \frac{dy}{y(y-1)} = dx \Rightarrow \frac{dy}{y-1} - \frac{dy}{y} = dx \Rightarrow \ln|y-1| - \ln|y| = x + c_1$$

$$\Rightarrow \ln \left| \frac{y-1}{y} \right| = x + c_1 \Rightarrow \left| \frac{y-1}{y} \right| = e^{c_1} e^x \Rightarrow \frac{y-1}{y} = ce^x$$

$$\Rightarrow 1 - \frac{1}{y} = ce^x \Rightarrow \frac{1}{y} = 1 - ce^x \Rightarrow y = \frac{1}{1-ce^x}$$

$$\text{Now } y(0) = -1 \Rightarrow -1 = \frac{1}{1-c} \Rightarrow 1-c = -1 \Rightarrow c = 2$$

Hence
$$y = \frac{1}{1-2e^x}$$

Ex: Logistic Equation:

$$y' = Ay - By^2$$

$y=0$ is a solution.

Otherwise it is a Bernoulli Equation with $a=2$

Let $u = y^{1-2} = y^{-1}$

$$u' = (-1)y^{-1-1}y' = -\frac{1}{y^2}y'$$

$$\Rightarrow u' = -\frac{1}{y^2}(Ay - By^2)$$

$$\Rightarrow u' = -Ay^{-1} + B$$

$$\Rightarrow u' + Au = B \quad (\text{Linear ODE})$$

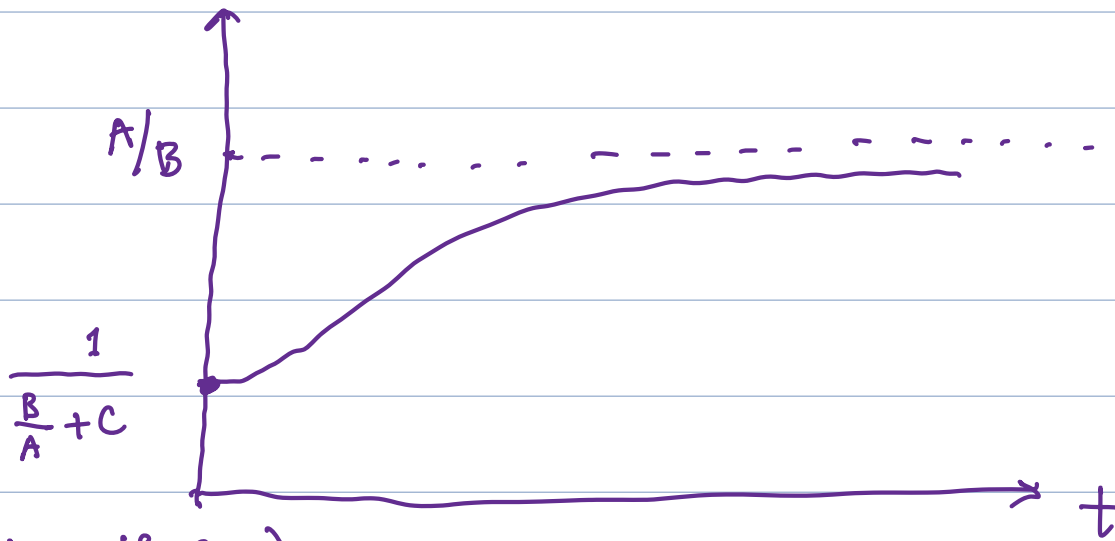
$$\text{I.F.} = e^{\int A dx} = e^{Ax}$$

$$\text{So, } u = e^{-Ax} \left[\int B e^{Ax} dx + C \right]$$

$$= e^{-Ax} \left[\frac{B}{A} e^{Ax} + C \right]$$

$$\Rightarrow \frac{1}{y} = \frac{B}{A} + C e^{-Ax}$$

$$\Rightarrow \boxed{y = \frac{1}{\frac{B}{A} + C e^{-Ax}}}$$



(In this graph $C > 0$)

Ex: Population Dynamics:

Let $y(t)$ be the population at time t .

Then the logistic Equation can be rewritten as:

$$y' = Ay - By^2$$

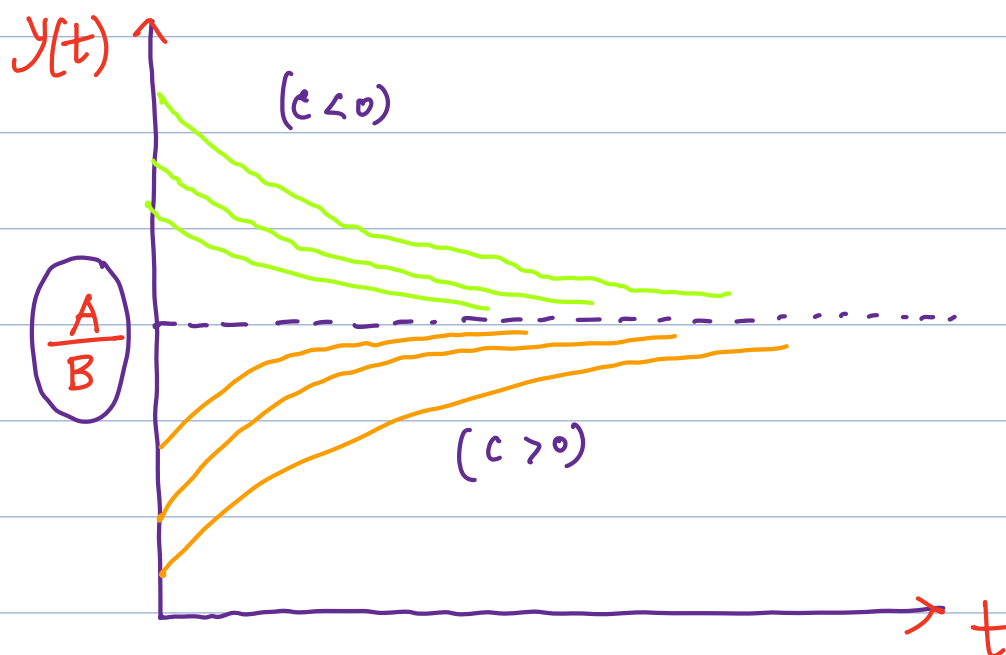
$$\Rightarrow y' = Ay \left(1 - \frac{B}{A}y\right) \quad A, B > 0$$

If $y < \frac{A}{B}$, then $y' > 0 \Rightarrow$ Population grows

If $y > \frac{A}{B}$, then $y' < 0 \Rightarrow$ Population decreases.

$\frac{A}{B}$: "equilibrium solution" "stable"

$$y(t) = \frac{1}{\frac{B}{A} + C e^{-At}}$$



Note: The term $-By^2$ acts as a 'braking term' that prevents the population from growing infinitely.

Note: For a small population

$$y' = Ay \quad (\text{Malthus Law})$$

(Particular case of Logistic Equation)

$$\Rightarrow y = \frac{1}{c e^{-At}} = \frac{1}{c} e^{At}$$

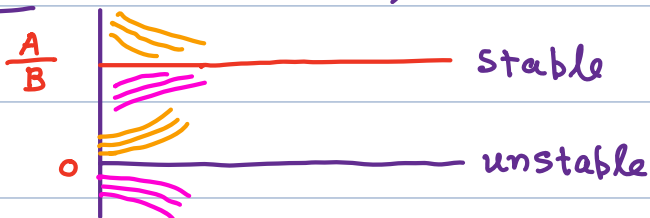
Autonomous ODE:

A general first order ODE $y' = f(x, y)$ in which $f(x, y)$ is just a function of y (i.e. x does not appear explicitly) is called an Autonomous ODE.
Thus it is of the form $y' = f(y)$

- Any solution of $f(y) = 0$ is called a critical point or equilibrium point. (At these points $y' = 0$ and so there is no change)
Note that a critical point gives a solution of the ODE since $\text{RHS} = f(y) = 0$ and $\text{LHS} = y' = 0$ (as y is a constant)

A critical point may be stable if solutions close to it for some t remain close to it for all further t or unstable if solutions initially close to it do not remain close as t increases.

Ex: $y' = Ay - By^2$, $A, B > 0$ Now $Ay - By^2 = 0 \Rightarrow y(A - By) = 0 \Rightarrow y = 0, y = \frac{A}{B}$

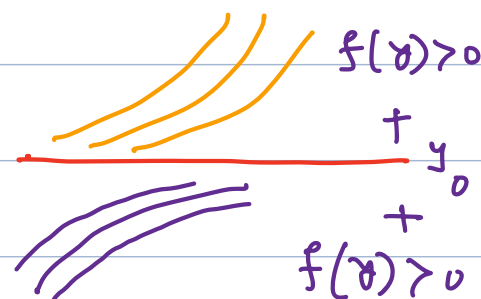
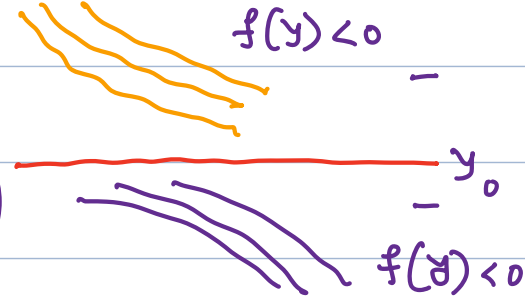
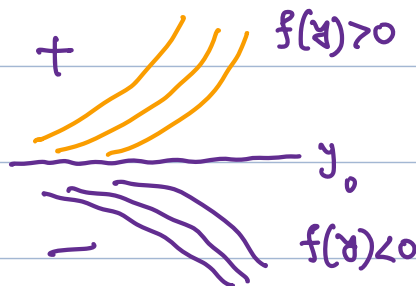
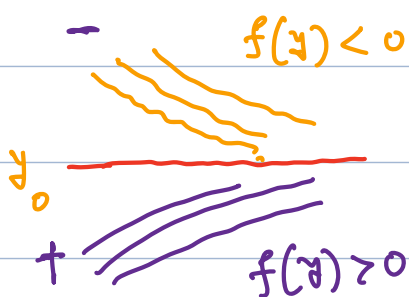
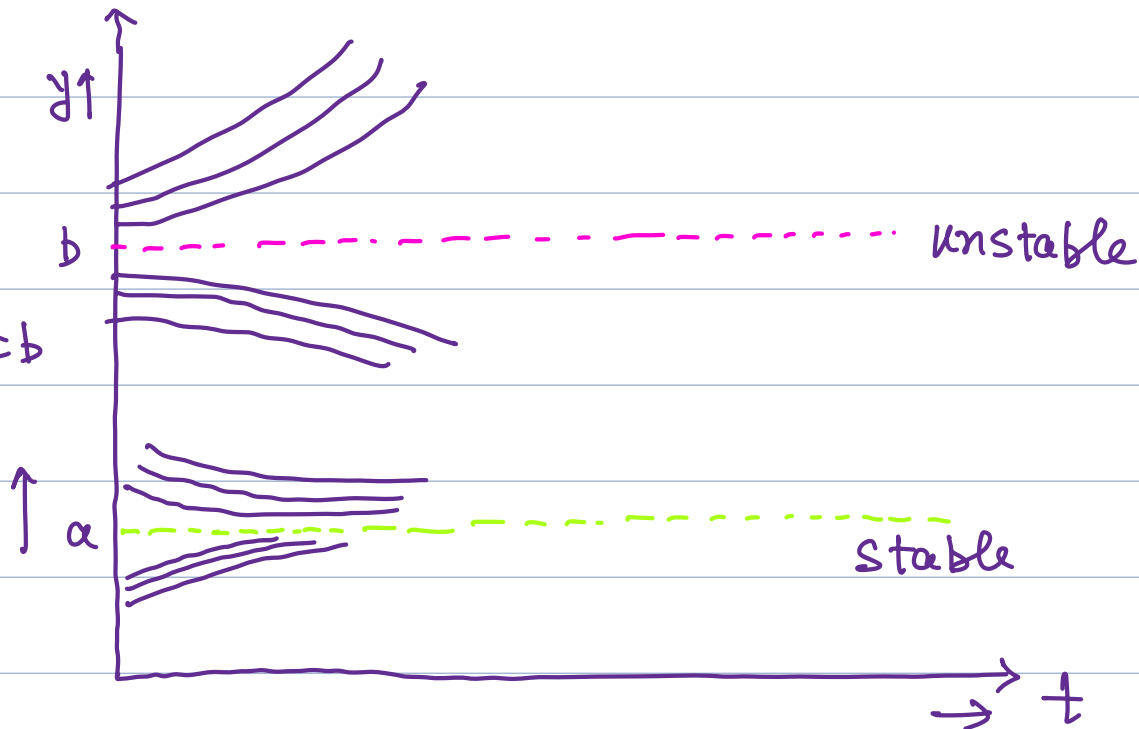


Suppose

$$y' = f(y)$$

and $f(y) = 0$

at $y = a$ & $y = b$



$f(y)$
doesn't
change sign
across y_0

Semi Stable

Ex: $\frac{dy}{dt} = -(y-10)^2(y-4)$ (Autonomous ODE)

Here $f(y) = -(y-10)^2(y-4)$

The critical points are 4, 10

At 4, $f(y)$ changes sign from '+' to '-'

So, 4 is a stable critical point.

At 10, $f(y)$ doesn't change its sign (It is '-' to '-')

So, 10 is a semi stable critical point.

Ex: $\frac{dy}{dt} = (y^3 - 8)(e^y - 1)$ (Autonomous ODE)
Here $f(y) = (y^3 - 8)(e^y - 1)$

Equilibrium points are 0 and 2.

At 0, $f(y)$ changes its sign from '+' to '-'

So, 0 is a stable equilibrium point

At 2, $f(y)$ changes its sign from '-' to '+'

So, 2 is a unstable equilibrium point.

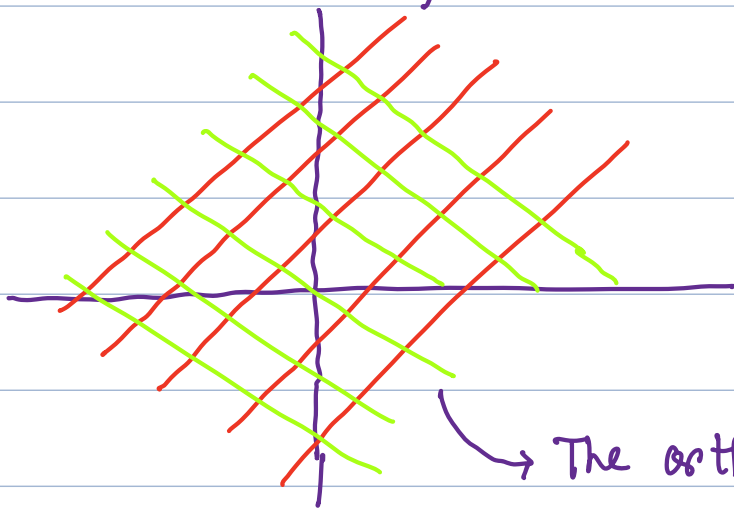
Orthogonal Trajectories

Let us consider the family of curves that are given by $G(x, y, c) = 0$

Question:

What is the family of curves that are orthogonal to the above family?

Ex: $y = \frac{1}{2}x + C$ is a given family of curves.



The orthogonal family is

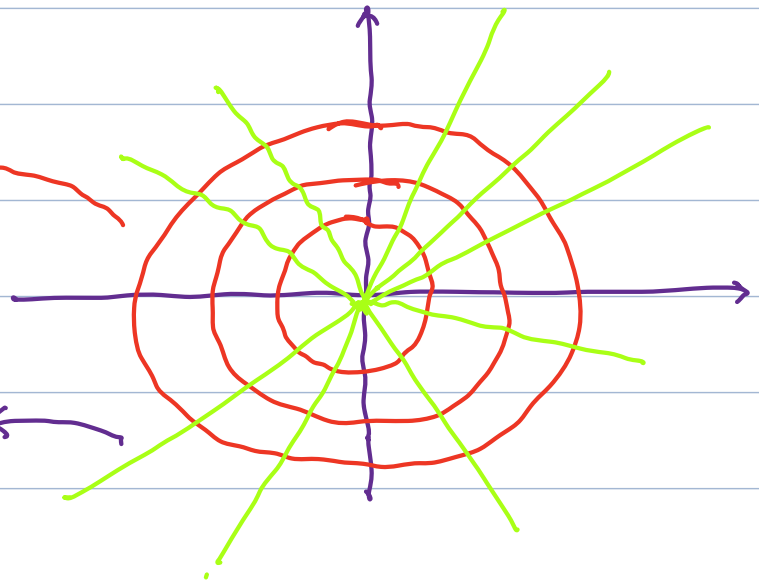
$$y = 2x + C_1$$

Ex:

$$x^2 + y^2 = C^2$$

The orthogonal family
is

$$y = C_1 x$$



Ex: Find the family of curves which are orthogonal to $\frac{x^2}{2} + y^2 = C$

Let us find the first order ODE whose solutions are $\frac{x^2}{2} + y^2 = C$

$$\Rightarrow \frac{2x}{2} + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{2y}$$

The orthogonal family will have slope

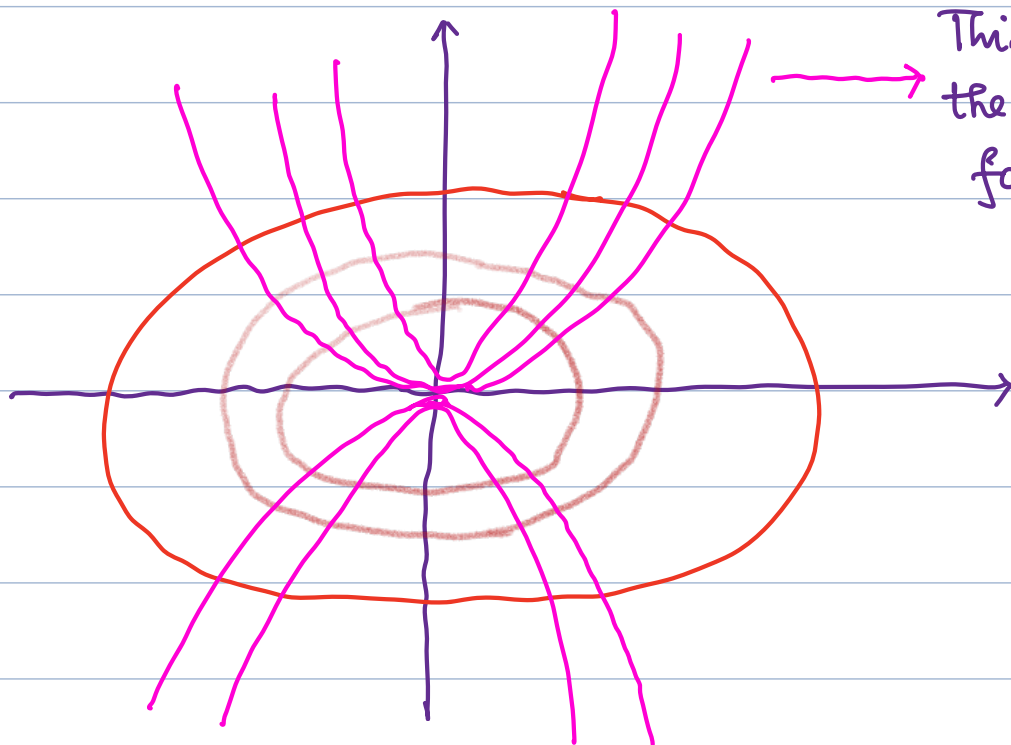
$$\frac{2y}{x}$$

So, $\frac{dy}{dx} = \frac{2y}{x}$

$$\Rightarrow \frac{dy}{y} = 2 \frac{dx}{x}$$

$$\Rightarrow \ln y = 2 \ln x + \ln C$$

$$\Rightarrow \ln y = \ln(Cx^2) \Rightarrow \boxed{y = Cx^2}$$



→ This is the orthogonal family of curves.

Ex:

$$xy = c$$

$$x \frac{dy}{dx} + y = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

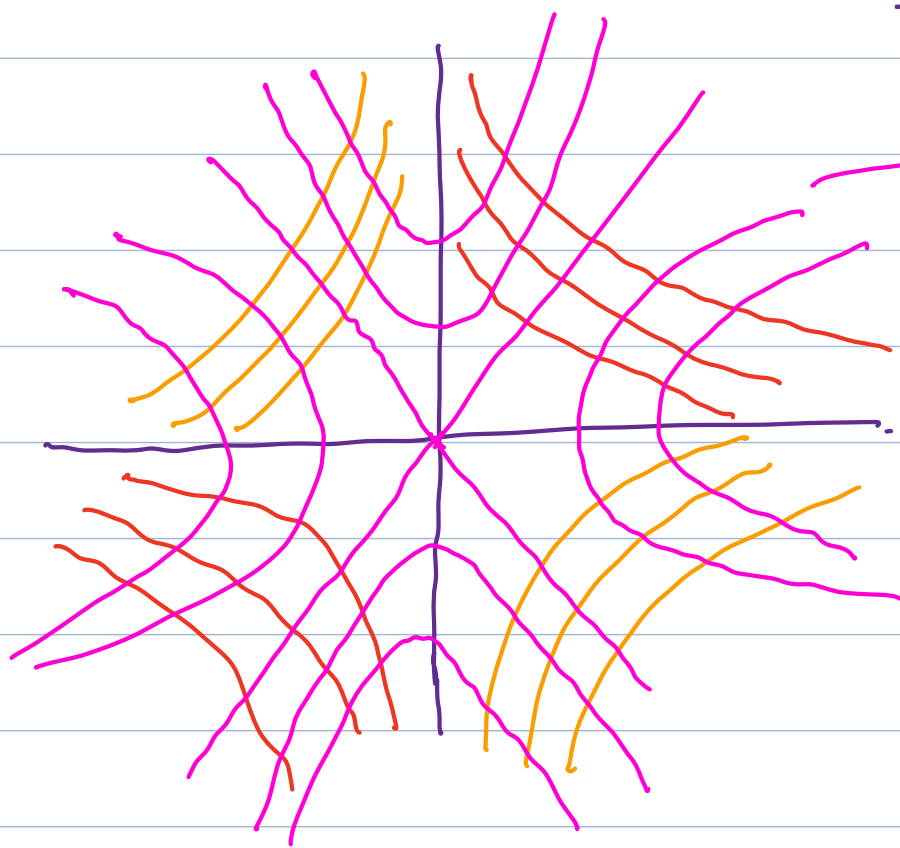
The orthogonal family will have slope

$$\frac{x}{y}$$

;

$$\frac{dy}{dx} = \frac{x}{y} \Rightarrow y dx = x dy$$

$$\Rightarrow \boxed{y^2 = x^2 + c}$$



This is the
orthogonal family
of curves

Steps to find orthogonal trajectories:

- (1) Find the first order ODE corresponding to the given family of curves by differentiating and eliminating the constant.
- (2) Find y' in terms of x and y from the

Obtained ODE, say $y' = f(x, y)$

(3) Step up a new ODE as $y' = -\frac{1}{f(x, y)}$

(4) Solve the ODE in step (3) to find the orthogonal family to the given family of curves.

Ex: $y = \sqrt{x+c} \Rightarrow y' = \frac{1}{2\sqrt{x+c}} = \frac{1}{2y}$

The orthogonal family will have slope $-2y$

$$\Rightarrow y' = -2y \Rightarrow \frac{dy}{y} = -2dx \Rightarrow \ln|y| = -2x + C_1$$

$$\Rightarrow \boxed{y = C e^{-2x}}$$